

● HARD

● ADVANCED MATH

● ANSWER KEY

Equivalent expressions

03

9 answers · Rationale-rich · Teach from doc

Q1**.2916**

Accept also: .2917, 7/24

The correct answer is $\frac{7}{24}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the right-hand side of the given equation, $\sqrt[3]{4^7}$, is equivalent to $4^{\frac{7}{3}}$. Thus, $4^{8c} = 4^{\frac{7}{3}}$. It follows that $8c = \frac{7}{3}$. Dividing both sides of this equation by 8 yields $c = \frac{7}{24}$. Note that 7/24, .2916, .2917, 0.219, and 0.292 are examples of ways to enter a correct answer.

Q2**B****B. 7**

Choice B is correct. Using the distributive property, the first given expression can be rewritten as $6x^2 + 3px + 24 - 16px - 64x + 24$, and then rewritten as $6x^2 + (3p - 16p - 64)x + 24$. Since the expression $6x^2 + (3p - 16p - 64)x + 24$ is equivalent to $6x^2 - 155x + 24$, the coefficients of the x terms from each expression are equivalent to each other; thus $3p - 16p - 64 = -155$. Combining like terms gives $-13p - 64 = -155$. Adding 64 to both sides of the equation gives $-13p = -91$. Dividing both sides of the equation by -13 yields $p = 7$.

Choice A is incorrect. If $p = -3$, then the first expression would be equivalent to $6x^2 - 25x + 24$. Choice C is incorrect. If $p = 13$, then the first expression would be equivalent to $6x^2 - 233x + 24$. Choice D is incorrect. If $p = 155$, then the first expression would be equivalent to $6x^2 - 2,079x + 24$.

Q3**A****A.** -8

Choice A is correct. The given expression is equivalent to $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$. By combining like terms, this can be written as $9x^4 + x^3 - 8x^2 - x$. This expression is in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. Therefore, the value of a is -8.

Choice B is incorrect. This may result from adding $-5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice C is incorrect. This may result from adding $5x^2$ and $-3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice D is incorrect. This may result from adding $5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Q4**A****A.** 4

Choice A is correct. If the given expression is equivalent to $x + b$, then $\frac{x^2 - c}{x - b} = x + b$, where x isn't equal to b . Multiplying both sides of this equation by $x - b$ yields $x^2 - c = (x + b)(x - b)$.

Since the right-hand side of this equation is in factored form for the difference of squares, the value of c must be a perfect square. Only choice A gives a perfect square for the value of c .

Choices B, C, and D are incorrect. None of these values of c produces a difference of squares.

For example, when 6 is substituted for c in the given expression, the result is $\frac{x^2 - 6}{x - b}$. The

expression $x^2 - 6$ can't be factored with integer values, and therefore $\frac{x^2 - 6}{x - b}$ isn't equivalent to $x + b$.

Q5**C**

C. $a^2 + 2ac + c^2$

Choice C is correct. Using the property that $\sqrt{x}\sqrt{y} = \sqrt{xy}$ for positive numbers x and y , with $x = (a + c)^3$ and $y = a + c$, it follows that $\sqrt{(a+c)^3} \cdot \sqrt{a+c} = \sqrt{(a+c)^4}$. By rewriting $(a + c)^4$ as $((a + c)^2)^2$, it is possible to simplify the square root expression as follows:
 $\sqrt{((a+c)^2)^2} = (a+c)^2 = a^2 + 2ac + c^2$.

Choice A is incorrect and may be the result of $\sqrt{(a+c)^3} \div \sqrt{(a+c)}$. Choice B is incorrect and may be the result of incorrectly rewriting $(a + c)^2$ as $a^2 + c^2$. Choice D is incorrect and may be the result of incorrectly applying properties of exponents.

Q6**D**

D. $3x^2 + 49x + 14b$

Choice D is correct. Since each choice has a term of $3x^2$, which can be written as $3xx$, and each choice has a term of $14b$, which can be written as $72b$, the expression that has a factor of $x + 2b$, where b is a positive integer constant, can be represented as $3x + 7x + 2b$. Using the distributive property of multiplication, this expression is equivalent to $3xx + 2b + 7x + 2b$, or $3x^2 + 6xb + 7x + 14b$. Combining the x -terms in this expression yields $3x^2 + 7 + 6bx + 14b$. It follows that the coefficient of the x -term is equal to $7 + 6b$. Thus, from the given choices, $7 + 6b$ must be equal to 7, 28, 42, or 49. Therefore, $6b$ must be equal to 0, 21, 35, or 42, respectively, and b must be equal to $\frac{0}{6}$, $\frac{21}{6}$, $\frac{35}{6}$, or $\frac{42}{6}$, respectively. Of these four values of b , only $\frac{42}{6}$, or 7, is a positive integer. It follows that $7 + 6b$ must be equal to 49 because this is the only choice for which the value of b is a positive integer constant. Therefore, the expression that has a factor of $x + 2b$ is $3x^2 + 49x + 14b$.

Choice A is incorrect. If this expression has a factor of $x + 2b$, then the value of b is 0, which isn't positive.

Choice B is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{21}{6}$, which isn't an integer.

Choice C is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{35}{6}$, which isn't an integer.

Q7**-419**

The correct answer is -419. It's given that the expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. Applying the distributive property to the given expression, $3x - 2319x + 6$, yields $3x19x + 3x6 - 2319x - 236$, which can be rewritten as $57x^2 + 18x - 437x - 138$. Combining like terms yields $57x^2 - 419x - 138$. Since this expression is equivalent to $ax^2 + bx + c$, it follows that the value of b is -419.

Q8**C**

C. $(x + a)(x - a)$

Choice C is correct. Factoring -1 from the second, third, and fourth terms gives $x^2 - c^2 - 2cd - d^2 = x^2 - (c^2 + 2cd + d^2)$. The expression $c^2 + 2cd + d^2$ is the expanded form of a perfect square: $c^2 + 2cd + d^2 = (c + d)^2$. Therefore, $x^2 - (c^2 + 2cd + d^2) = x^2 - (c + d)^2$. Since $a = c + d$, $x^2 - (c + d)^2 = x^2 - a^2$. Finally, because $x^2 - a^2$ is the difference of squares, it can be expanded as $x^2 - a^2 = (x + a)(x - a)$.

Choices A and B are incorrect and may be the result of making an error in factoring the difference of squares $x^2 - a^2$. Choice D is incorrect and may be the result of incorrectly combining terms.

Q9**D**

D. $\frac{45}{k}$

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $hx + kx + j$. The expression $hx + kx + j$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + jh + kx + kj$. Therefore, $hx^2 + jh + kx + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $\frac{-45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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